

Mixed-Precision Randomized Sketching for Scaled Dot-Product Attention

Ján Kovačovský

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Faculty of Mathematics and Physics, Charles University

Instructor: H. V. Le

Abstract. We propose a mathematically focused study of scaled dot-product attention under two simultaneous approximations: randomized feature sketching and mixed-precision arithmetic. The exact logit operator $d^{-1/2}QK^T$ is replaced by the sketched operator $d^{-1/2}QSS^TK^T$, where S is a random sketching matrix of smaller column dimension. The report isolates three levels of perturbation: the deterministic error in the logits, the induced perturbation of the rowwise softmax map, and the additional finite-precision error introduced when the sketched computation is carried out in low precision. The main analytical result is a clean error decomposition showing that the total output error is controlled by the sum of a sketching term and a rounding term. The accompanying experiments are designed as a validation of this decomposition rather than as a large-scale engineering benchmark.

Keywords: mixed-precision arithmetic, randomized numerical linear algebra, attention, transformers, sketching

1 Proposal

The purpose of this report is to study one analytically transparent fragment of the transformer architecture [1]: scaled dot-product attention. The report is intentionally not about full transformer training. Instead, it isolates a single nonlinear operator and examines how two approximations interact:

- (1) a randomized finite-rank compression motivated by randomized numerical linear algebra [2] [3];
- (2) a mixed-precision realization motivated by modern low-precision training practice [4].

This focus keeps the project aligned with the mathematical language of the lecture notes: dimension reduction, kernel-style matrix constructions, and operator perturbation. It also places the work in the neighborhood of efficient-attention models such as Linformer, Nyströmformer, and Performer [5] [6] [7], while remaining much narrower in scope.

The guiding question is the following.

Idea 1 (Research question). When the attention logits are first compressed by a randomized sketch, can the subsequent computation be carried out in mixed precision with ↴

only a lower-order loss of accuracy? ┐

The intended contribution of the report is modest but precise: a deterministic error decomposition, together with numerical experiments that test whether the finite-precision term is indeed subordinate to the sketching term in relevant parameter regimes.

2 Operator-Theoretic Setup

Let $\mathcal{H}_d := \mathbb{R}^d$ and $\mathcal{H}_n := \mathbb{R}^n$, both endowed with their standard Euclidean inner products. We identify matrices in $\mathbb{R}^{n \times d}$ with Hilbert–Schmidt operators

$$Q, K, V : \mathcal{H}_d \rightarrow \mathcal{H}_n.$$

Their Hilbert–Schmidt norm is denoted by $\|\cdot\|_{\text{HS}}$, while $\|\cdot\|_2$ denotes the spectral norm on linear operators.

The exact logit operator is

$$\mathcal{L}(Q, K) := d^{-1/2} Q K^* \in \mathcal{L}(\mathcal{H}_n),$$

and the rowwise softmax map is denoted by

$$\Sigma : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}.$$

Accordingly, the attention output is the nonlinear operator

$$\mathcal{A}(Q, K, V) := \Sigma(\mathcal{L}(Q, K))V.$$

Now let $S \in \mathbb{R}^{d \times s}$ with $s < d$. The associated rank- s positive semidefinite operator on $\mathcal{H}_d$ is

$$P_S := S S^*.$$

The sketched logit operator is then

$$\tilde{\mathcal{L}}_S(Q, K) := d^{-1/2} Q P_S K^*,$$

and the corresponding sketched attention output is

$$\tilde{\mathcal{A}}_S(Q, K, V) := \Sigma(\tilde{\mathcal{L}}_S(Q, K))V.$$

Thus the basic approximation step is the replacement of the identity on $\mathcal{H}_d$ by the finite-rank operator P_S .

3 Analytical Core

The analysis begins with a stability property of the rowwise softmax map.

Proposition 2 (Rowwise softmax is Hilbert–Schmidt Lipschitz). Let $L, M \in \mathbb{R}^{n \times n}$. Then

$$\|\Sigma(L) - \Sigma(M)\|_{\text{HS}} \leq \|L - M\|_{\text{HS}}.$$

Consequently, for every $V \in \mathbb{R}^{n \times d}$,

$$\|\Sigma(L)V - \Sigma(M)V\|_{\text{HS}} \leq \|L - M\|_{\text{HS}} \|V\|_2.$$

Proof. Let $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the ordinary softmax map on vectors. Its Jacobian at $x \in \mathbb{R}^n$ is

$$D\sigma(x) = \text{Diag}(p) - pp^T, \quad p := \sigma(x).$$

For any $z \in \mathbb{R}^n$ one has

$$z^T D\sigma(x) z = \sum_{i=1}^n p_i z_i^2 - \left(\sum_{i=1}^n p_i z_i \right)^2 \leq \sum_{i=1}^n p_i z_i^2 \leq \|z\|_2^2.$$

Hence $\|D\sigma(x)\|_2 \leq 1$ for all x , and the mean value theorem yields

$$\|\sigma(x) - \sigma(y)\|_2 \leq \|x - y\|_2.$$

Applying this estimate row by row gives

$$\|\Sigma(L) - \Sigma(M)\|_{\text{HS}}^2 = \sum_{i=1}^n \|\sigma(\ell_i) - \sigma(m_i)\|_2^2 \leq \sum_{i=1}^n \|\ell_i - m_i\|_2^2 = \|L - M\|_{\text{HS}}^2,$$

where ℓ_i and m_i are the rows of L and M . The second assertion follows from the submultiplicativity estimate

$$\|XV\|_{\text{HS}} \leq \|X\|_{\text{HS}} \|V\|_2.$$

□

The next proposition isolates the deterministic sketching error.

Proposition 3 (Deterministic sketching bound). Let $U \subseteq \mathcal{H}_d$ be a subspace containing all row vectors of Q and K . Assume that P_S is an ϵ -approximate identity on U , in the sense that

$$\|(I - P_S)u\|_2 \leq \epsilon \|u\|_2 \quad \text{for all } u \in U.$$

Then

$$\|\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2.$$

Consequently,

$$\|\mathcal{A}(Q, K, V) - \tilde{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2 \|V\|_2.$$

Proof. Since

$$\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K) = d^{-1/2} Q(I - P_S) K^*,$$

we obtain

$$\|\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \leq \frac{1}{\sqrt{d}} \|Q(I - P_S)\|_{\text{HS}} \|K^*\|_2.$$

□

↳ Every row of Q lies in U , so the approximate-identity assumption implies

$$\|Q(I - P_S)\|_{\text{HS}} \leq \epsilon \|Q\|_{\text{HS}}.$$

Using $\|K^*\|_2 = \|K\|_2$ proves the first estimate. The second follows from the previous proposition. $\square$

This reduces the sketching analysis to a single geometric requirement on the projector P_S . In the randomized setting, the role of randNLA is to choose S so that this requirement holds with high probability on a subspace that captures the geometry of the rows of Q and K .

The final step is to separate sketching error from finite-precision error.

Proposition 4 (Error splitting for mixed-precision attention). Let $\hat{\mathcal{L}}_S(Q, K)$ denote the logit matrix produced by a finite-precision implementation of the sketched computation, and define

$$\hat{\mathcal{A}}_S(Q, K, V) := \Sigma(\hat{\mathcal{L}}_S(Q, K))V.$$

Then

$$\|\mathcal{A}(Q, K, V) - \hat{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \left(\|\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K)\|_{\text{HS}} + \|\tilde{\mathcal{L}}_S(Q, K) - \hat{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \right) \|V\|_2.$$

Proof. Apply the previous Lipschitz estimate with

$$L = \mathcal{L}(Q, K), \quad M = \hat{\mathcal{L}}_S(Q, K),$$

and then use the triangle inequality

$$\|L - M\|_{\text{HS}} \leq \|L - \tilde{\mathcal{L}}_S(Q, K)\|_{\text{HS}} + \|\tilde{\mathcal{L}}_S(Q, K) - M\|_{\text{HS}}.$$

$\square$

Remark 5 (Interpretation). The first term in the bound is a purely sketching term; the second is a purely numerical term. In the intended mixed-precision regime, the numerical term should be lower order. This is the precise statement that the experiments are meant to test. $\lrcorner$

4 Numerical Experiments

The experiments probe the decomposition of Proposition 3 directly. For each combination of parameters (n, d, s) and each precision configuration we compute three logit matrices:

- (1) $\mathcal{L}(Q, K)$: exact unsketched logits in `float64` (reference);
- (2) $\tilde{\mathcal{L}}_S(Q, K)$: exact sketched logits in `float64`;
- (3) $\hat{\mathcal{L}}_S(Q, K)$: sketched logits in the target precision.

The sketching term is $\|\mathcal{L} - \tilde{\mathcal{L}}_S\|_{\text{HS}}$ and the rounding term is $\|\tilde{\mathcal{L}}_S - \hat{\mathcal{L}}_S\|_{\text{HS}}$, both measured in Hilbert–Schmidt norm. The same decomposition is recorded for the attention output.

We test three precision configurations: **full_fp32** (all stages in float32), **full_fp16** (all stages in float16), and **mixed_fp16** (storage in float16, accumulation and softmax in float32).

The sketch $S \in \mathbb{R}^{d \times s}$ is a scaled Gaussian matrix with i.i.d. $\mathcal{N}(0, 1/s)$ entries. We sweep $n \in \{64, 128, 256\}$, $d \in \{32, 64\}$, $s \in \{8, 16, 24, 32\}$, and average over three seeds.

Table 1 reports the HS errors of the logits for $d = 64$; the output errors follow the same pattern and are omitted for brevity. The sketching term is 3–4 orders of magnitude larger than the rounding term under **full_fp32**, and 1–2 orders larger under **full_fp16** and **mixed_fp16**. In all tested regimes the total error $\|\mathcal{L} - \hat{\mathcal{L}}_S\|_{\text{HS}}$ is dominated by the sketching term, confirming the hypothesis of Proposition 3.

Configuration	s	E_{sk}	E_{rd}	$E_{\text{sk}}/E_{\text{rd}}$
full_fp32	8	6.23e+02	9.92e-05	6.3e+06
	16	4.73e+02	9.79e-05	4.8e+06
	24	3.49e+02	7.86e-05	4.4e+06
	32	3.10e+02	7.42e-05	4.2e+06
full_fp16	8	6.23e+02	3.70e-01	1.7e+03
	16	4.73e+02	2.51e-01	1.9e+03
	24	3.49e+02	2.12e-01	1.6e+03
	32	3.10e+02	1.82e-01	1.7e+03
mixed_fp16	8	6.23e+02	3.38e-01	1.8e+03
	16	4.73e+02	2.30e-01	2.1e+03
	24	3.49e+02	1.95e-01	1.8e+03
	32	3.10e+02	1.67e-01	1.9e+03

Table 6 / Hilbert–Schmidt errors of the logit matrix ($d = 64$, averaged over 3 seeds). Sketching error $E_{\text{sk}} := \|\mathcal{L} - \tilde{\mathcal{L}}_S\|_{\text{HS}}$; rounding error $E_{\text{rd}} := \|\tilde{\mathcal{L}}_S - \hat{\mathcal{L}}_S\|_{\text{HS}}$; ratio $E_{\text{sk}}/E_{\text{rd}}$.

Two observations stand out. First, the ratio $E_{\text{sk}}/E_{\text{rd}}$ is consistently above 10^3 even for naive float16, and above 10^6 for float32. Second, the mixed-precision configuration achieves a rounding error comparable to naive float16 (within a factor of 2) while the sketching error is identical by construction. This is consistent with the intuition that the rounding term is governed by the reduced dynamic range after sketching, not by the precision of the outer accumulation.

5 Outlook

The present report deliberately studies only the feature-sketch model $QK^T \approx QSS^TK^T$. This is sufficient for a mathematically clean first analysis. The natural extensions are Nyström-type approximations and random-feature approximations of the softmax kernel, which would connect the project more directly with the efficient-attention literature [6] [7].